AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT131
Discrete Mathematics

Midterm Examination (MTA)
Fall Semester 2025/2026

Date:18 /Nov. /2025

Number of Exam Pages: 3 pages (including this cover sheet)	**Time Allowed: 90 Minutes**

Instructions:

- Total Marks: **30**
- This exam consists of **4** Questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Q–1: [5×3 marks] Choose the correct answer:

1.1: The negation of $(P \wedge Q) \longrightarrow R$ is

a) $\neg (P \wedge Q) \longrightarrow \neg R$
b) $\neg R \longrightarrow \neg (P \wedge Q)$
c) $R \longrightarrow (P \wedge Q)$
d) $P \wedge Q \wedge \neg R$
e) None of the above

1.2 Find a counterexample for the statement were $U = \{3, 5, 7, 9\}$ is the universal set:

$$"\forall x, x + 3 \geq 7"$$

a) 3 is a counterexample.
b) The statement is true; hence no counterexample exists.
c) 9 is the only counterexample.
d) 5 is a counterexample
e) None of the above

1.3: Let $f: R - \left\{\frac{3}{5}\right\} \rightarrow R$ be defined by $f(x) = \frac{3x+2}{5x-3}$. Then

a) $f^{-1}(x) = f(x)$
b) $f^{-1}(x) = -f(x)$
c) $(f \circ f)\, x = -x$
d) $f^{-1}(x) = \frac{1}{19} f(x)$
e) None of the above

1.4: The quotient and remainder when –5 is divided by 6 is

a) $q = 5, r = 6$
b) $q = 0, r = 6$
c) $q = 0, r = 5$
d) $q = -1, r = 1$
e) None of the above

1.5: For the given Arithmetic sequence find the first negative term?

60, 55, 50, 45, ...

a) -1
b) -2
c) -3
d) -5
e) None of the above

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Each question in below worth 15 marks. Answer all the following:

Q–2:

a) **[4 marks]** Determine the truth values of $\forall x \left((x \geq 1) \leftrightarrow (x \geq \sqrt{x})\right)$, where the domain of discourse is the set of natural numbers, justify your answer.

b) **[2+2 marks]** Write the negation of the following statements:

i. It is neither cold nor dry.
ii. If it is rainy, then we go to the movies.

c) **[7 marks]** Show that the statement $\neg((P \wedge \neg Q) \rightarrow P \vee Q)$ is a contradiction using Truth table.

Q–3:

a) **[2+2+2 marks]** Let $U = \{n \in Z^* \mid n \leq 12\}$ be the universal set, $A = \{n \in U \mid n \text{ is odd}\}$, and $B = \{n \in U \mid n \text{ is a multiple of } 4\}$.

i. List the elements of U, A, and B.
ii. Find $(A \cap B) \cup U$.
iii. Find the power set of B.

b) **[6 marks]** Find a, b and c if:

$$\begin{cases} a = 43 \; div \; 6 \\ a + b = -51 \; mod \; 6 \\ a + c = 64 \; mod \; 8 \end{cases}$$

c) **[3 marks]** Find the value of the following sum: $\sum_{j=1}^{5}(10j)$.

Q–4:

a) **[8 marks]** Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ Find $A^{[2]}$, if possible.

b) **[7 marks]** Encrypt the message "$DEAR\ DOCTOR$" using the function $f(p) = (p + 10) \; mod \; 26$.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

End of questions

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